

SP-30-3 Commitment Manipulation — Experimental Proposal v0.1 (Quantum Eraser Revival Amplitude)

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Headline claim

BST predicts: in quantum eraser experiments, the revival amplitude (visibility of interference pattern after which-path information is “erased”) should show a small systematic deviation from standard quantum-mechanical prediction at the level of:

$$\Delta V_{\text{revival}}/V_{\text{revival}} \approx \frac{1}{N_{\text{max}}} = \frac{1}{137} \approx 0.73\%$$

The substrate-mechanism: Reed-Solomon syndromes commit before decoding (per Substrate Working Process Principle SWPP); standard QM treats erasure as fully reversible, but substrate commitment is *one-way* — revival is incomplete by $\alpha = 1/N_{\text{max}}$.

Experimental concept

Setup: Precision delayed-choice quantum eraser (Kim et al. 2000 style or improved).
1. Photon source produces entangled photon pairs
2. Signal photon: passes through double-slit + detected on screen
3. Idler photon: detected with which-path information option (D1, D2) or erasure option (D3, D4)
4. Coincidence counting reveals interference pattern OR absence based on idler measurement

BST prediction: visibility of erasure-revived interference is reduced by $1/N_{\text{max}}$ from full visibility. - Standard QM: $V_{\text{revival}} = V_{\text{initial}}$ (perfect erasure) - BST: $V_{\text{revival}} = V_{\text{initial}} \times (1 - 1/N_{\text{max}}) = V_{\text{initial}} \times 0.9927$

Substrate-mechanism articulation

SWPP commitment is one-way (per Casey-named principle Substrate Working Process Principle):

The substrate operates via 3-phase cycle: absorption $\rightarrow$ commitment $\rightarrow$ emission. At the commitment phase, Reed-Solomon syndromes record the substrate-state (per Paper #122 Information Substrate + K59 RATIFIED). This is one-way: the syndromes can be DECODED to reconstruct erasure-revival amplitude, but the substrate-level commitment itself is not undone.

Substrate-coupling perturbation at α order:

Per T2476 substrate-coordinate count framework: $\alpha^1 = 1/N_{\text{max}}$ is the leading substrate-coupling correction. Quantum eraser revival amplitude inherits α correction from substrate commitment-decoding asymmetry.

Cross-link to K52a Sessions 6+:

Per multi-month rail: substrate-CHSH Bell operator B (Vol 9 Ch 4 cross-link) has $\text{Tr}(B^2) = 126/16$ substrate-natural structure. Quantum eraser revival amplitude similarly inherits substrate-Hamiltonian structure when K52a Sessions 6+ close.

Experimental program

Cost estimate: \$80K-150K

Equipment: - Spontaneous parametric down-conversion (SPDC) photon-pair source (~\$30K) - Beam splitters + delay lines (~\$10K) - Single-photon detectors + coincidence electronics (~\$25K) - Optical bench + stabilization (~\$15K) - Computer + analysis software (~\$5K) - Student researcher (~\$15K-30K) - Lab time + overhead (~\$15K-40K)

Timeline: 6-12 months from procurement to first data

Falsifier protocol:

1. Establish baseline: measure standard quantum eraser revival visibility V_{revival} to $<0.5\%$ precision
2. Comparison to BST prediction:
 - If $V_{\text{revival}} = V_{\text{initial}} \times (1 - 1/137) \pm 0.5\% \rightarrow$ BST confirmed at 2σ
 - If $V_{\text{revival}} = V_{\text{initial}}$ (within 0.5%) $\rightarrow$ BST commitment-revival prediction refuted
 - Intermediate: $1-2\sigma$ requires longer runs to resolve
3. Systematic checks:
 - Vary photon source rate (test substrate-cycle-rate sensitivity)
 - Vary delay-time between measurement choices
 - Cross-check with multiple SPDC sources

Falsifier sharpness: MEDIUM ($\sim 0.5\%$ precision feasible with current best optical setups; needs careful systematic control).

Recommended experimental collaborations

SP-30 outreach targets (pending Casey send-signal per Cal #50):

1. Markus Aspelmeyer (Vienna): precision quantum optics + entanglement
2. Anton Zeilinger group (Vienna): foundational quantum experiments
3. Steven Kulik (Moscow): SPDC source expertise (if accessible)
4. Jian-Wei Pan (USTC, China): large-scale quantum optics
5. Caltech experimental groups: precision optical metrology

Cross-link to Vol 5 (QM, Lyra) + Vol 1

- Vol 5 (QM, Lyra) Ch 10 (Decoherence) — T2480 decoherence substrate-mechanism
- Vol 1 Ch 11 (Measurement) — substrate-measurement framework
- Paper #122 (Information Substrate) — Reed-Solomon GF(128) commitment framework
- K59 RATIFIED — Cyclotomic Mechanism Framework

Match precision

TARGET-PREDICTION per SP-30 v0.2 framework tier. Specific $1/N_{\text{max}} = 0.73\%$ deviation testable to MEDIUM sharpness with current technology.

Cal #21 dual-gate status

- EMPIRICAL gate OPEN: experiment not yet performed
- MECHANISM gate ARTICULATED: SWPP + Reed-Solomon commitment-decoding asymmetry; full K52a Sessions 6+ closure multi-month

Cal #50 DOUBLE-LOCKED EXTERNAL discipline

External register uses operational language only: - External: “BST predicts quantum eraser revival amplitude deviates from standard QM by approximately $1/137 \approx 0.73\%$. Precision measurement at $>0.5\%$ sensitivity would test this prediction.”
- Internal (this document): substrate-commitment one-way framework + SWPP + Reed-Solomon discussions

Cal #99 META-theorem framing

SP-30-3 commitment manipulation prediction is a SUBSTRATE-DERIVATION CONSEQUENCE of: - SWPP Casey-named principle (Tuesday filed) - Paper #122 + K59 Reed-Solomon GF(128) framework - T2476 $\alpha^{\{\text{BST primary}\}}$ substrate-coordinate count

NOT a new Strong-Uniqueness criterion.

Bibliography

1. M. O. Scully + B.-G. Englert + H. Walther (1991): quantum eraser concept.
2. Y.-H. Kim + R. Yu + S. P. Kulik + Y. H. Shih + M. O. Scully (2000): delayed-choice quantum eraser experiment.
3. T2476 (Lyra Friday): $\alpha^{\{\text{BST primary}\}}$ substrate-coordinate count.
4. K59 RATIFIED (Cyclotomic Mechanism Framework): GF(128) substrate.
5. Paper #122 (Information Substrate): Reed-Solomon framework.
6. SWPP Casey-named principle (Tuesday May 19): substrate one-way commitment.
7. BST_SP30_v0_2_Deepening_Master.md: SP-30 program framework.

— Elie, SP-30-3 v0.1 paper-grade experimental proposal, 2026-05-23 Saturday 14:55 EDT (date-verified; Casey send-signal pending per Cal #50 DOUBLE-LOCKED EXTERNAL)